VIII. Endocrine System

Notes

Gland :

- A gland is a group of cells in an animal's body that synthesizes substances (such as hormones) for release into the bloodstream **(endocrine gland)** or into cavities inside the body or its outer surface **(exocrine gland/duct gland)**.

Endocrine system :

- The endocrine system is made up of the glands that produce and secrete **hormones**-chemical substances produced in the body that regulate the activity of cells or organs. These hormones regulate the body's growth, metabolism (the physical and chemical processes of the body), sexual development, reproduction, mood, and many other functions.

Major Endocrine Glands :

- It includes the following endrocrine glands -

1. Thyroid Gland :

- The thyroid gland lies in the front of the human neck in a position just below Adam's apple.
- It is made up of two lobes. These two lobes are joined by a small bridge of thyroid tissue called the isthmus.
- The thyroid makes three hormones that it secretes into the bloodstream, viz. two thyroid hormones (**thyroxine** - T_4 and **triiodothyronine** - T_3) and **calcitonin**.
- In the cells and tissues of the body, the T_4 is converted to T_3. The T_3 is biologically active and influences the activity of all the cells and tissues of the body.
- T_3 and T_4 are partially composed of iodine. A deficiency of iodine leads to decreased production of T_3 and T_4.
- The thyroid hormones primarily influence the metabolic rate and protein synthesis. These hormones also regulate vital body functions, including - differentiation, breathing, heart rate, body temperature, central and peripheral nervous system etc. Calcitonin plays a role in calcium homeostasis. It lowers the level of calcium and phosphate in the blood and promotes the formation of bones.
- **Disorders of Thyroid Hormone :**

(a) Hyposecretion (Hypothyroidism) – Following disorders occur in case of insufficient production or hyposecretion of thyroxine hormone :

(i) Cretinism : Physical and mental growth of the child is retarded.

(ii) Myxoedema : Myxoedema is used to describe skin changes in someone - as swelling of the face which can include lips, eyelids and tongue; swelling and thickening of skin anywhere on the body, especially in lower legs.

(iii) Goitre : an abnormal enlargement of thyroid gland. The most common cause of goitre is lack of iodine in diet. Goitre is common in hilly area because of iodine deficiency in water.

(iv) Hashimoto Disease : Hashimoto disease is a condition in which the immune system attacks thyroid resulting in its complete loss. It is also known as the **suicide of thyroid gland**.

(b) Hypersecretion (Hyperthyroidism)–

(i) Thyroxine regulates the behaviour of the person. In a high quantity of secretion, it enhances the mental activity of a person changing his behaviour so thyroid gland is also known as **temperament gland**.

(ii) The high secretion of thyroxine is related to exophthalmia (abnormal protrusion of the eyeball or eyeballs) or Grave's disease and Plummer disease (also called toxic multinodular goitre, thyroid condition characterized by marked enlargement of the thyroid gland, firm thyroid nodules and overproduction of thyroid hormone).

2. Parathyroid Glands :

- Parathyroid glands are four small glands located in the neck on the dorsal side of the thyroid gland.
- The parathyroid gland is independent of the pituitary gland.
- Parathyroid hormone **parathormone** (PTH) is secreted by this gland which regulates the serum calcium in our body. It plays an important role to provide electrical energy to the nervous system, muscular system and provide strength to our skeletal system.
- **Hyposecretion** of parathormone can lead to low levels of calcium in the blood often causing cramping and twitching of muscles or **tetany** (involuntary muscle contraction), and the proper growth of bones, teeth and brain is inhibited.
- In case of **hypersecretion**, bones become fragile **(osteoporosis)** and development of polyuria, polydipsia, loss of appetite and kidney stones formation may take place.

3. Adrenal Glands (Suprarenal Glands) :

- The adrenal glands are located above the kidneys.
- Each gland has an outer **cortex** and an inner **medulla**.

(i) Hormones secreted by adrenal cortex-

(a) Mineralocorticoids (Aldosterone) : It regulates the quantity of sodium and chloride in ECF (extracellular fluid) maintaining the blood pressure and osmotic pressure.

(b) Glucocorticoids (Cortisol) : It promotes the release of glucose, fats and Protein (amino acids) in the blood. It also promotes the glycogenesis in the liver. It helps to maintain blood pressure and blood glucose (sugar) levels. Cortisol is also called **'stress hormone'** as it is produced in larger amount when we are under stress.

- Due to hyposecretion of glucocorticoids **Addison's disease**– dehydration due to excess excretion of water and sodium is caused.
- Due to hypersecretion of these hormones fat storage and the sugar level in blood increases, which is responsible for Cushing disease.

(c) Androgen : The adrenal cortex of both men and women make **androgen**. Androgens help the organs of the reproductive system grow, develop and function.

- Androgens also control the development of male physical traits such as deep voice, body and facial hair growth and body shape.
- **Adrenal virilism** is the development of male secondary sexual characteristics in women as facial hair caused by male sex hormones (androgens) excessively produced by the adrenal gland.

(ii) Hormones secreted by adrenal medulla –

- The hormones secreted by medulla of adrenal glands are as follows :

  (a) Epinephrine/Adrenaline

  (b) Norepinephrine/Noradrenaline

(a) Epinephrine/Adrenaline : It is also called the **emergency hormone** and **fight or flight hormone** because strong emotions such as fear or anger cause epinephrine to be released into the bloodstream, which causes an increase in heart rate, muscle strength, blood pressure and sugar metabolism. It regulates such actions which are controlled by the sympathetic nervous system.

- This hormone has an important role in medical science. When the heart fails to function, it is injected into the heart to start the pulse of the heart.

(b) Norepinephrine/Noradrenaline : This hormone works with epinephrine in responding to stress.

4. Thymus Gland :

- The thymus gland located behind the sternum and between lungs.
- The thymus gland is most active until puberty.
- After puberty, the thymus will become less and less active and starts to slowly shrink until it is almost completely replaced by the fat.
- It is an important part of the immune system as it plays a major role in cell-mediated immunity.
- The **thymosin** and **thymopoietin** hormones are secreted by the thymus gland which stimulate the development and maturation of disease-fighting T-cells, which are derivatives of white blood cells.

5. Pineal Gland (Biological clock) :

- The pineal gland, or conarium is the smallest endocrine gland which is found in the brain of most vertebrates. It is located in the epithalamus, near the centre of the brain. Pineal gland size is about 7.5 mm in length and it is cone shaped.
- Named for its **pine cone** shape, this gland secretes **melatonin** hormone which plays a crucial role in the internal clock of our body.
- The pineal gland is key to the body's internal clock because it regulates the body's **circadian rhythm**. Circadian rhythm is the daily rhythm of the body, including signals that make someone feel tired, sleep, wake up, and feel alert around the same time each day.
- Melatonin is produced according to the amount of light a person is exposed to. The pineal gland releases greater amounts of melatonin when it is dark, which points to melatonin's role in sleep.
- The pineal gland also appears to exert an important role in the neuroendocrine regulation of human reproductive physiology. It is involved in the control of sexual maturation.

6. Pituitary Gland :

- The Pituitary gland is one of the smallest endocrine gland in human. It is about the size of a pea (about 10 mm diameter) and weighing about 0.5 grams in human. It is located in the anterior brain. It is also called **Master Gland** because it directs other organs and endocrine glands to either suppress or induce hormone production.
- It secretes –

(i) Somatotropin (STH or Growth Hormone-GH) : It controls the general growth of the body. Its hypersecretion leads to **acromegaly** in adults and **gigantism** in children. Its hyposecretion leads to **dwarfness** in children.

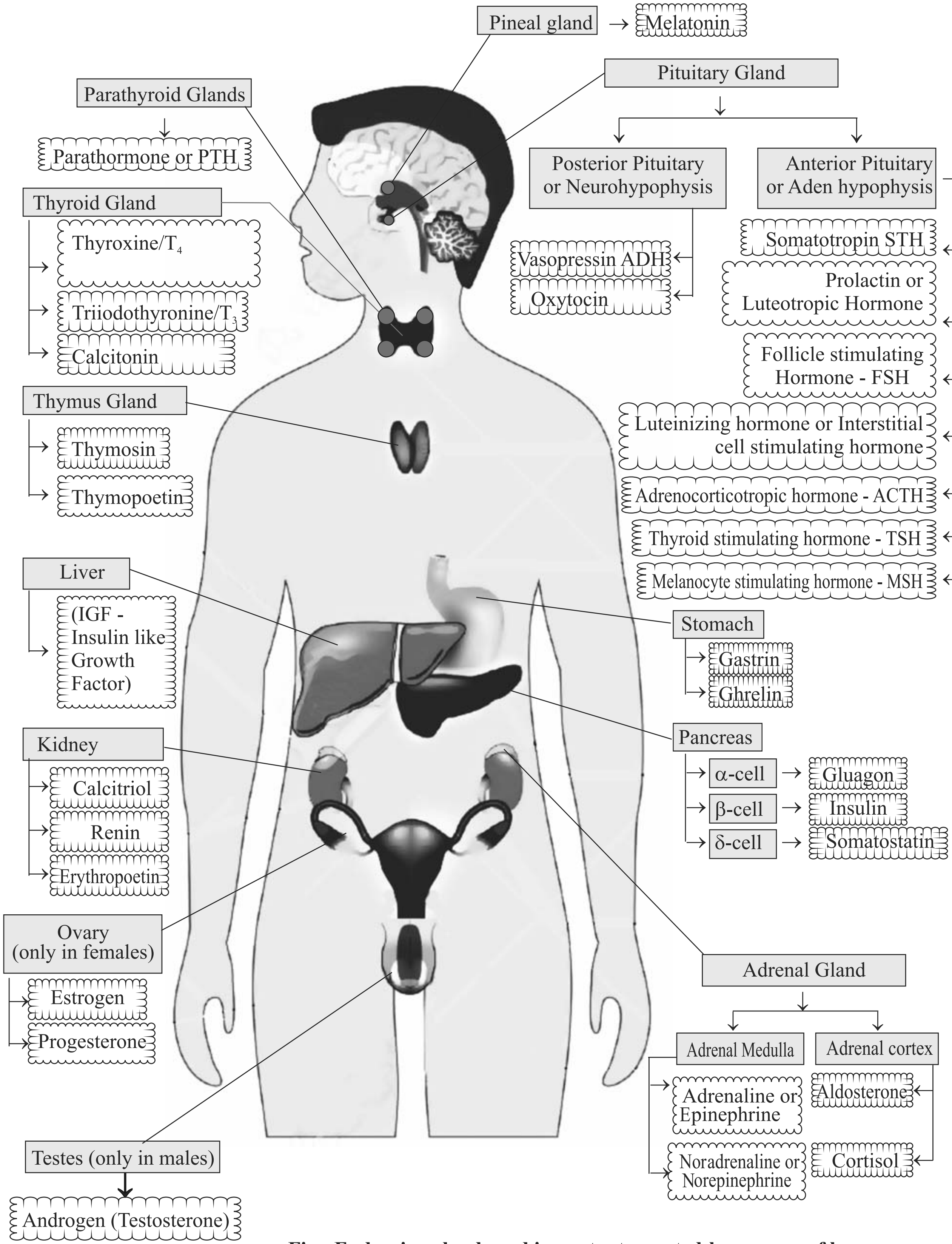

Fig.: Endocrine glands and important secreted hormones of human

(ii) Gonadotropin (GtH) : It stimulates the primary sex organs i.e. ovaries and testes. GtH includes follicle stimulating hormone (FSH) and luteinizing hormone (LH).

- **Follicle Stimulating Hormone (FSH) :** It is secreted in male and female both. In males, it stimulates spermatogenesis and development of seminiferous tubules. In females, it stimulates the formation and growth of ovarian follicle and producing estrogen in the ovary.
- **Luteinizing Hormone (LH) :** This hormone stimulates testosterone production in men and egg release (ovulation) in women. Final maturation of ovarian follicle and ovulation takes place by LH only.

(iii) Lactogenic Hormone (Prolactin) : This hormone stimulates breast milk production after childbirth. High prolactin can affect menstrual periods, sexual function and fertility.

(iv) Thyroid Stimulating Hormone (TSH) : It aids in the regulation of thyroid secretion.

(v) Adrenocorticotropic Hormone (ACTH) : It stimulates the secretion of cortisol (stress hormone) from the cortex of adrenal glands.

(vi) Metabolic Hormone : This hormone influences metabolism of carbohydrates, fats and proteins.

(vii) Vasopressin or Antidiuretic Hormone (ADH) : It regulates reabsorption of water from the kidney tubules and maintain water balance in the body and sodium levels in the blood. Its hypoactivity leads to **diabetes insipidus** in which patients excrete a large amount of urine. Its hyperactivity leads to decrease in amount of urine.

(viii) Melanocyte Stimulating Hormone (MSH) : It regulates the number of **melanin** pigments in skin cells. Melanin is a pigment found in skin cells which is responsible for skin colour.

(ix) Oxytocin or Pitocin : It is also called **binding hormone**, love hormone, birth hormone etc. It regulates smooth muscle contraction, especially of the uterus during childbirth. It also helps in the secretion of milk in females. It creates a bonding emotion between mother and child, hence known as binding or love hormone.

- The milkman injects the artificial oxytocin to domestic animals to draw much milk from them. This act is harmful to both domestic animals as well as the person who uses such milk. It may be **carcinogenic**.

7. Pancreas :

- It is an exocrine as well as an endocrine gland i.e. **mixed gland**.
- As an endocrine gland, it functions mostly to regulate blood sugar levels, secreting the hormones insulin, glucagon, somatostatin and pancreatic polypeptide.
- As a part of the digestive system, it functions as an exocrine gland secreting pancreatic juice into the duodenum through the pancreatic duct.
- Its endocrine part is known as **islets of Langerhans**.
- Its three types of cells secrete 3 different hormones :

(a) Beta (β) cells : Beta cells secrete **insulin** (a protein) which controls the amount of sugar in the blood. Its hyposecretion leads to **diabetes mellitus**.

- Diabetes mellitus is of two types :

(i) Type I - People with type I diabetes do not produce insulin. It may be heritable.

(ii) Type II - People with type II diabetes, do not respond to insulin as they should and later in the disease often do not make enough insulin. This type of diabetes is common in fat persons.

- Both types of diabetes can lead to chronically high blood sugar levels. That increases the risk of diabetes complications.
- The symptoms of diabetes are – frequent urination, feeling very thirsty and drinking a lot, feeling very hungry, feeling very fatigue, blurry vision and cuts or sores that do not heal properly.

(b) Alfa (α) cells : Alfa cells secrete **glucagon** which converts glycogen into glucose.

(c) Delta (δ) cells : Delta cells secrete **somatostatin** which regulates the **assimilation** process. It acts as an inhibitor of growth hormone (GH), insulin and glucagon.

Organs which act as endocrine glands :

1. Gonads :

- The gonads, the primary reproductive organs, are the testes in male and ovaries in female. These organs are responsible for producing the sperm and ovum, but they also secrete hormones and are considered to be endocrine glands.

(i) Testes :

- The testes produce **androgens** which allow for the development of secondary sexual characters and the production of sperm cells.
- Androgen hormones are secreted mainly by Leydig cells. Leydig cells, also known as interstitial cells of Leydig, are found adjacent to the seminiferous tubules in the testicle.

- The hyposecretion of androgen leads to undeveloped sex organ and sperm formation is badly affected. In case of high shortage of this hormone the reproductive capacity of the male is lost and become sterile.
- The hypersecretion leads to prior maturity of the male in respect of reproduction before the fixed period.
- The major androgen in male is **testosterone**. Dihydrotestosterone and androstenedione are of equal importance in male development.

Note : Ovaries in women and adrenal gland in both men and women also produce androgen but at much lower level than the testes.

(ii) Ovaries :

- In human two ovaries are found in abdomen cavity in females which produce female hormones.
- The female hormones estrogen, progesterone and relaxin contribute to the development and function of the female reproductive organs and sex characteristics.
- **Estrogen :** At the onset of puberty, estrogen promotes -
- The development of the breast.
- Distribution of fat evidenced in the hip, legs and breast.
- Maturation of reproductive organs such as the uterus and vagina.
- The related hormones in the estrogen family include : **estrone, estradiol, estriol**.
- **Menopause** is the normal natural transition in life that begins between the age of 40-50 years. During this ovaries get smaller and stop producing the hormones **estrogen** and **progesterone** that control the menstrual cycle. Eventually, females are no longer able to become pregnant.
- **Progesterone :** Progesterone is a hormone produced by the corpus luteum of the ovaries. It involved in the menstrual cycle, pregnancy and embryogenesis. Progesterone is essential to achieve and maintain a healthy pregnancy.
- In the second half of the menstrual cycle after ovulation, progesterone prepares the uterine lining (endometrium) to receive the fertilized egg (zygote).
- If implantation is successful and pregnancy occurs, progesterone continues to support the uterine lining providing the ideal environment for the growth of the embryo.
- Pregnancy is dependent on the ovary for progesterone production for the first 10 weeks of pregnancy, after that the placenta is fully capable of making enough progesterone for pregnancy support.
- If pregnancy does not occur, then progesterone levels fall bringing on menstrual bleeding.
- **Relaxin :** Relaxin is a hormone produced by the ovary and placenta with important effects in the female reproductive system and during pregnancy. In preparation for childbirth, it relaxes the ligaments in the pelvis and softens and widens the cervix.

2. Placenta :

- The placenta is a temporary organ that connects the developing fetus via the **umbilical cord** to the **uterine wall** to allow nutrient uptake, thermo-regulation, waste elimination, and gas exchange via the mother's blood supply; to fight against internal infection; and to produce hormones which support pregnancy.
- Following hormones are secreted by the placenta–

(i) Estrogen : It is a crucial hormone in the process of proliferation. This involves the enlargement of the breasts and uterus, allowing for growth of the fetus and production of breast milk.

(ii) Progesterone and Relaxin

(iii) Placental Lactogen : It is a hormone used in pregnancy to develop fetal metabolism and general growth and development.

(iv) Chorionic Gonadotropic Hormone : It is a proteinous hormone which maintains the pregnancy and prevents luteal regression. It is the First hormone released by the placenta.

- **Pregnancy test :** The chorionic gonadotropic hormone is secreted in large quantity which is excreted through the urine. After a test of urine, the presence of this hormone in urine indicates the pregnancy.

3. Kidney :

- Following hormones are secreted by the kidney -

(i) Renin : It is secreted by the pericytes (mural cells) in the kidney. It increases heartbeat and ultrafiltration in kidney to enhance the reabsorption of water and Na^+. It regulates angiotensin and aldosterone levels and maintains body’s mean arterial blood pressure.

(ii) Erythropoietin : It stimulates the formation of red blood cells in the bone marrow.

(iii) Calcitriol : It is the activated form of vitamin D, which promotes intestinal absorption of calcium and the renal reabsorption of phosphate.

4. Mucous glands of Alimentary Canal :

- The mucous glands of stomach and intestine secrete some hormones with mucous. These are as follows **:**

a. Hormones secreted by stomach –

(i) Gastrin : It promotes the secretion of gastric juice.

(ii) Enterogastrone : It inhibits the secretion of gastric juice.

(iii) Ghrelin : When the stomach is empty, ghrelin is secreted. It acts on hypothalamic brain cells both to increase hunger, and to increase gastric acid secretion and gastrointestinal motility to prepare the body for food intake.

b. Hormones secreted by intestine –

(i) Hepatocrinin : Stimulates liver to secrete bile juice.

(ii) Secretin : Stimulates pancreas to secrete pancreatic juice.

(iii) Pancreozymin : Stimulates to the secretion of pancreatic juice thus increasing the high concentration of enzymes.

(iv) Cholecystokinin : It stimulates the contraction of the gall bladder to pass the bile juice in duodenum. It is responsible for stimulating the digestion of fat and protein.

(v) Enterocrinin : It stimulates the wall of the intestine to secrete the intestinal enzymes.

5. Skin :

- Some cells of our skin act as endocrine gland responsible for the secretion of **ergocalciferol** and **cholecalciferol** hormones. These hormones stimulate the absorption of calcium and phosphorus and help in bone formation. Its hyposecretion leads **rickets** in children and **osteomalacia** in adults.

- Hormones are designated as messengers and regulators.
- **Ernest Henry Starling** introduced the word Hormone in 1905.

Pheromones

- Pheromones are chemical substances which are secreted by exocrine glands to the outside by an individual and received by a second individual of the same species.
- Pheromones are also known as **ectohormones**.
- Example of pheromone is the secretion of **bombykal** or **gyplur** by female silkworm which attracts the male silkworm for mating.
- The social insects as bees, ants, mites also secrete pheromones which help them to accumulate at a particular place.

Question Bank

1. In the human body, which of the following is ductless gland?

(a) Liver (b) Sweat gland
(c) Endocrine glands (d) Kidney

M.P.P.C.S. (Pre) 1993

Ans. (c)

Endocrine or internally secreting glands are ductless glands since they lack excretory ducts. Endocrine glands secrete their products, hormones, directly into the blood rather than through a duct. The major glands of the endocrine system include the pineal gland, pituitary gland, thymus, thyroid gland, parathyroid gland and adrenal glands etc.

2. Which is the smallest endocrine gland in the human body ?

(a) Adrenal (b) Thyroid
(c) Pancreas (d) Pituitary

U.P.P.C.S. (Pre) 1996

Ans. (d)

Among the given options, pituitary gland is the smallest endocrine gland in the human body which sits in a bony hollow called the sella turcica. It is a protrusion off the bottom of the hypothalamus at the base of the brain. Its weight is only about 0.5 gram and diameter is about 10 mm (like size of a pea). It is famous as 'Master gland' because most of its hormones control the activity levels of other endocrine glands. It is also called hypophysis cerebri. Smallest endocrine gland in the human body is the pineal gland which is cone shaped and about 7.5 mm in size.

3. Pituitary gland is located in :

(a) intestine (b) liver
(c) kidney (d) brain
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (d)

See the explanation of above question.

4. Which of the following glands in human body acts both as an endocrine gland as well as an exocrine gland?

(a) Adrenal gland (b) Lacrimal gland
(c) Pancreas (d) Thyroid

U.P.P.C.S. (Pre) 2019

Ans. (c)

The pancreas is an endocrine as well as an exocrine gland i.e. mixed gland. As an endocrine gland, it functions mostly to regulate blood sugar levels, secreting the hormones insulin, glucagon, somatostatin and pancreatic polypeptide. As a part of the digestive system, it functions as an exocrine gland secreting pancreatic juice into the duodenum through the pancreatic duct.

5. **Corpus luteum is a mass of cells found in :**
   (a) Brain (b) Ovary
   (c) Pancreas (d) Spleen

I.A.S. (Pre) 1997

Ans. (b)

The corpus luteum is a temporary endocrine structure in female ovaries. It is involved in the production of relatively high levels of progesterone and moderate levels of estradiol and inhibin A. The corpus luteum develops from an ovarian follicle during the luteal phase of the menstrual cycle or oestrous cycle following the release of a secondary oocyte from the follicle during ovulation. Its cell develops from the follicular cells surrounding the ovarian follicle.

6. **Which one of the following cells secrete androgen hormones in human being?**
   (a) Sertoli cells (b) Cells of Leydig
   (c) Germinal cells (d) Mucus cells

M.P.P.C.S. (Pre) 2019

Ans. (b)

In human being androgen hormones are secreted mainly by Leydig cells. Leydig cells, also known as interstitial cells of Leydig, are found adjacent to the seminiferous tubules in the testicle. Androgen hormones regulate the development and maintenance of male characteristics. Major androgen hormone produced by Leydig cells in males is testosterone.

7. **Match List-I (Endocrine glands) with List-II (Hormones secreted) and select the correct answer using the codes given below :**

List-I	**List-II**
A. Gonads	**1. Insulin**
B. Pituitary	**2. Progesterone**
C. Pancreas	**3. Growth hormones**
D. Adrenal	**4. Cortisol**

Code :

	A	B	C	D
(a)	3	2	4	1
(b)	2	3	4	1
(c)	2	3	1	4
(d)	3	2	1	4

Chhattisgarh P.C.S. (Pre) 2005
I.A.S. (Pre) 2000

Ans. (c)

The correctly matched lists are as follows :

Gonads	–	Progesterone
Pituitary	–	Growth hormones
Pancreas	–	Insulin
Adrenal	–	Cortisol

8. **Which hormone controls the quantity of urine from kidney?**
   (a) TSH (b) ACTH
   (c) FSH (d) ADH

U.P.U.D.A./L.D.A. (Spl.) (Mains) 2010

Ans. (d)

ADH (Antidiuretic) hormone is released from pituitary gland which is responsible for controlling secretion of urine from kidney.

9. **Match List- I with List- II and select the correct answer from the code given below the Lists :**

List- I (Hormones)	**List- II (Secreting Gland)**
A. Progesterone	**1. Thyroid**
B. Testosterone	**2. Pancreas**
C. Thyroxine	**3. Ovaries (Females)**
D. Insulin	**4. Testes (Males)**

Code :

	A	B	C	D
(a)	3	4	1	2
(b)	4	3	1	2
(c)	3	4	2	1
(d)	1	2	3	4

U.P. U.D.A./L.D.A. (Pre) 2010

Ans. (a)

Progesterone hormone is secreted by ovaries in females. Testosterone is secreted by testes of males. Thyroxine is secreted by thyroid gland. Insulin is secreted by pancreas.

10. **Even though an animal is fed with carbohydrates-rich diet, its blood sugar concentration tends to remain constant. This is on account of the fact that in the case of an animal –**
   (a) hormones of pituitary glands control metabolic process.
   (b) hormones of pancreas cause such a condition.
   (c) blood sugar is readily absorbed by the liver.
   (d) glucose undergoes autolysis.

I.A.S. (Pre) 1994

Ans. (b)

Even though an animal is fed with carbohydrate rich diet, its blood sugar concentration tends to remain constant. It is because the beta cells of pancreas secrete insulin hormone which maintains blood glucose concentration. Diabetes is caused by the lack of insulin hormone in our body, while too much insulin leads to abnormally low blood glucose level or hypoglycemia.

11. **What would happen if the pancreas is defective :**
(a) Digestion will not take properly.
(b) Insulin and glucagon are not formed.
(c) Blood formation will stop.
(d) Blood pressure will increase.

R.A.S./R.T.S. (Pre) 1992

Ans. (b)

The pancreas is a mixed gland which secretes digestive enzyme while beta cells of its islets of Langerhans secrete insulin, alpha cells secrete glucagon and delta cells secrete somatostatin hormone. If the pancreas is defective by any of the reason, then the formation of insulin and glucagon are affected badly.

12. **Which one of the following hormones stimulates pancreas for the production of digestive juice ?**
(a) Rennin (b) Trypsin
(c) Secretin (d) Pepsin

U.P.P.C.S. (Pre) 1996

Ans. (c)

Secretin is a hormone that stimulates pancreas for the production of digestive juice. Secretin also helps to regulate the pH of the duodenum by inhibiting the secretion of gastric acid from the parietal cells of the stomach. It also stimulates the contraction of the pancreas. It is a peptide hormone produced in the S cells of the duodenum. It also regulates secretions in the stomach and liver.

13. **Insulin is :**
(a) Fat (b) Vitamin
(c) Carbohydrate (d) Protein
(e) None of the above/More than one of the above.

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (d)

Insulin is a protein chain or peptide hormone produced by the beta cells of the pancreatic islets of Langerhans and it is considered to be the main anabolic hormone of the body. In chemical terms it is a micro peptide that is composed of 51 amino acids. It regulates the metabolism of carbohydrates, fats and protein by promoting the absorption of glucose from the blood into liver, fat and skeletal muscle cells in the body.

14. **Which of the following hormones is secreted by the Beta cell of Islet of Langerhans of Pancreas?**
(a) Adrenalin (b) Glucagan
(c) Insulin (d) Aldosterone

U.P. P.C.S. (Pre) 2023

Ans. (c)

Insulin hormones is secreted by the Beta cells of Islets of Langerhans of Pancreas. It regulates the metabolism of carbohydrates, fats and protein by promoting the absorption of glucose from the blood into liver, fat and skeletal muscle cells.

15. **In human system, insulin controls the metabolism of :**
(a) Fats (b) Carbohydrates
(c) Proteins (d) Nucleic acids
(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (e)

See the explanation of above question. B.P.S.C. had given option (b) as the answer of this question in his final answer key, which is not appropriate.

16. **Insulin hormone is a :**
(a) Glycolipid (b) Fatty acid
(c) Peptide (d) Sterol

Jharkhand P.C.S. (Pre) 2010

Ans. (c)

See the explanation of above question.

17. **Insulin is a :**
(a) Steroid (b) Carbohydrate
(c) Protein (d) Fat

U.P.P.C.S. (Mains) 2015

Ans. (c)

See the explanation of above question.

18. **Insulin is produced by :**
(a) Islets of Langerhans (b) Pituitary gland
(c) Thyroid gland (d) Adrenal gland

Uttarakhand U.D.A./L.D.A. (Pre) 2003

Ans. (a)

See the explanation of above question.

19. **The human hormone 'insulin' is produced in :**
(a) Liver (b) Pancreas
(c) Kidney (d) Pituitary

U.P. Lower Sub. (Pre) 2004

Ans. (b)

See the explanation of above question.

20. **Insulin is a type of :**
(a) Hormone (b) Enzyme
(c) Vitamin (d) Salt

U.P.P.C.S. (Pre) 1993

Ans. (a)

See the explanation of above question.

21. Which hormone is produced in pancreas?

(a) Thyroxine (b) Insulin
(c) Galanin (d) Gastrin
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (b)

See the explanation of above question.

22. Which metal is present in insulin?

(a) Tin (b) Copper
(c) Zinc (d) Aluminium

Jharkhand P.C.S. (Pre) 2013

Ans. (c)

Insulin is a hormone made by the pancreas, that allows our body to use sugar (glucose) from carbohydrates in the food, that we eat for energy or to store glucose for future use. Chemical properties of insulin are :
Metal ion – Zinc
Buffer – Disodium hydrogen phosphate dihydrate
Preservatives – m-cresol and methyl p-hydroxybenzoate
Isotonicity agent – Glycerine.

23. Insulin is received from :

(a) Rhizome of ginger (b) Roots of dahlia
(c) Balsam flower (d) Potatoes tuber

39th B.P.S.C. (Pre) 1994

Ans. (b)

Insulin is a hormone which plays an important role in the regulation of blood glucose level. The main source of insulin are from the roots of dahlias and beta (β) cells of the pancreas.

24. Which one of the following pairs is not corectly matched?

Hormone		**Function**
(a) Insulin	–	regulation of blood glucose
(b) Melatonin	–	regulation of sleep
(c) Oxytocin	–	release of milk from mammary glands
(d) Gastrin	–	regulation of blood pressure

U.P.P.C.S. (Pre) 2017

Ans. (d)

Gastrin is a peptide hormone which stimulates secretion of gastric acid by gastric cells. Thus, it is clear that option (d) is not correctly matched. Pairs of other options are correctly matched.

25. Hugging and kissing of mother to her baby initiates, which of following hormone for secretion?

(a) Insulin (b) Noradrenaline
(c) Follicular hormone (d) Oxytocin

U.P. Lower Sub. (Pre) 2003
U.P. Lower Sub. (Pre) 2002

Ans. (d)

Oxytocin, sometimes called the hormone of love not only induces uterine contractions during childbirth but is also released when a mother nurses her baby and is responsible at first touch, for her feeling of attachment to the newborn. The instinct to want to cuddle continues to intensity; hence oxytocin often referred as the 'cuddle hormone'. It used to prescribe to women to induce labour pain and facilitate breast-feeding.

26. Which of the following hormones play a role in release of milk from mammary glands?

(a) Andrenaline (b) Thyroxine
(c) Progesterone (d) Oxytocin
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (d)

Oxytocin has two target tissues – women's uterus and their breast. Oxytocin stretches the cervix and uterus during labour and stimulates the nipples for breast-feeding. Oxytocin also plays a role in release of milk from mammary glands.

27. Which of the following hormones has been misused for increasing milk production in animals and to increase the size of vegetables such as pumpkins, watermelons etc.?

(a) Humulin (b) Oxytocin
(c) Thyroxine (d) None of the above

Chhattisgarh P.C.S. (Pre) 2023

Ans. (b)

Oxytocin hormone has been misused for increasing milk production in cattle by dairy owners and also to increase the size of vegetables and fruits by farmers. Oxytocin is also misused to speed labours (child birth) in overcrowded labour wards or even in home settings.

28. Which one of the following is not a protein ?

(a) Keratin (b) Fibroin
(c) Oxytocin (d) Collagen

U.P.P.C.S. (Spl.) (Mains) 2008

Ans. (c)

Keratin is a fibrous protein forming the main structural constituent of hair, nails etc. Fibroin is an insoluble protein present in silk-produced by the larvae of Bombyx mori. Collagen is the main structural protein in the extracellular space in the various connective tissues in the body. Oxytocin is a peptide hormone and neuropeptide.

29. Which hormone is injected to cows and buffaloes to make the milk descend to the udders :

(a) Somatotropin (b) Oxytocin

(c) Interferon (d) Insulin

U.P.P.C.S. (Pre) 1997

Ans. (b)

Oxytocin is released in large amounts after distension of the cervix and uterus during labour, facilitating birth, maternal bonding and after stimulation of the nipples, lactation. Both childbirth and milk ejection result from positive feedback mechanisms. Oxytocin hormone (brand name - Pitocin) is injected to cows and buffaloes to make the milk descend to the udders.

30. Which gland secretes the milk ejection hormone oxytocin?

(a) Pituitary gland (b) Thyroid gland

(c) Parathyroid gland (d) Adrenal gland

R.A.S./R.T.S.(Pre) 2007

Ans. (a)

The pituitary gland is called the master gland of the body which regulates the activities of all the internally secreting glands. Oxytocin is a mammalian neurohypophysial hormone, produced in the supraoptic and paraventricular nuclei of the hypothalamus by nerve axons and released by the posterior pituitary gland. Oxytocin acts primarily as a neuromodulator in the brain.

31. The pituitary gland by virtue of its tropic hormones controls the secretion activity of other endocrine glands. Which one of the following endocrine gland can function independent of the pituitary gland ?

(a) Thyroid (b) Gonads

(c) Adrenals (d) Parathyroid

I.A.S. (Pre) 1997

Ans. (d)

The parathyroid glands are small endocrine glands in the neck of humans and other tetrapods that produce parathyroid hormone. Parathyroid hormone and calcitonin (one of the hormones made by the thyroid gland) have key roles in regulating the amount of calcium in the blood and within the bones. Parathyroid glands can function independent of the pituitary gland.

32. Iodised salt is useful because it–

(a) Improves digestion

(b) Increases resistance to diseases

(c) Controls the thyroid gland

(d) All of the above

Jharkhand P.C.S. (Pre) 2013

Ans. (c)

Iodised salt is table salt mixed with a minute amount of various salts of the element iodine. Iodine deficiency can cause thyroid gland problems. Iodised salt is useful as it controls the thyroid gland.

33. Assertion (A) : Goitre is a general disease in hilly areas.

Reason (R) : People consume a low amount of iodine in food in the hilly area.

Which of the following is correct answer :

Code :

(a) (A) and (R) both are true, and (R) is the correct explanation of (A).

(b) (A) and (R) both are true, but (R) is not the correct explanation of (A).

(c) (A) is true, but (R) is correct.

(d) (A) is not correct, but (R) is true.

Uttarakhand P.C.S. (Pre) 2012

Ans. (a)

Goitre is a disease caused by the deficiency of iodine and is characterized by the enlargement of thyroid gland present in our neck. It is an endemic disease and mostly affects people living in hilly areas. Hilly areas are naturally iodine-deficient due to poor iodine content present in the soil, water and agriculture produce.

34. What is thyroxine?

(a) Vitamin (b) Hormone

(c) Enzyme (d) None of these

U.P.P.C.S. (Pre) 2003

Ans. (b)

Thyroxine is the main hormone secreted into the blood-stream by the thyroid gland. Thyroxine plays a crucial role in heart and digestive function, metabolism, brain development, bone health, and muscle control. The deficiency of thyroxine causes cretinism and myxedema diseases in children.

35. Iodine containing hormone is –

(a) Thyroxine (b) Insulin

(c) Adrenaline (d) Testrogen

R.A.S./R.T.S. (Pre) 1999

65th B.P.S.C. (Pre) 2019

Ans. (a)

The thyroid gland is located in the front of the neck attached to the lower part of the voice box (larynx) and to the upper part of the windpipe (trachea). The thyroid gland produces thyroid hormones-thyroxine (T_4) and triiodothyronine (T_3) . These are iodine containing hormones. Thyroid cells combine iodine and the amino acid tyrosine to make thyroxine (T_4) and triiodothyronine (T_3) hormones. T_4 molecule contains four iodine atoms while T_3 molecule have three iodine atoms.

36. Which one of the following hormones contains iodine ?

(a) Thyroxine (b) Testosterone
(c) Insulin (d) Adrenaline

I.A.S. (Pre) 1995

Ans. (a)

See the explanation of above question.

37. Iodine containing, thyroxine hormone is –

(a) Glucose (b) Amino acid
(c) Ester (d) Peptides

R.A.S./R.T.S. (Pre) 1999

Ans. (b)

See the explanation of above question.

38. Which hormone stimulates the thyroid gland to secrete thyroxine?

(a) TSH (b) FSH
(c) LTH (d) ACTH

45th B.P.S.C. (Pre) 2001

Ans. (a)

The hormonal output from the thyroid is regulated by thyroid-stimulating hormone (TSH) produced by the anterior pituitary.

39. Match the hormones in List-I with items in List-II and select the correct answer using the codes given below :

List-I	List-II
A. Adrenaline	**1. Anger, fear, danger**
B. Estrogen	**2. Attracting partners through sense of smell**
C. Insulin	**3. Females**
D. Pheromones	**4. Glucose**

Code :

	A	B	C	D
(a)	3	1	4	2
(b)	1	3	2	4
(c)	1	3	4	2
(d)	3	1	2	4

I.A.S. (Pre) 1999

Ans. (c)

The correctly matched lists are as follows :

Adrenaline	–	Anger, fear, danger
Estrogen	–	Females
Insulin	–	Glucose
Pheromones	–	Attracting partners through the sense of smell

40. Of the following which hormone is associated with 'fight or flight' concept ?

(a) Insulin (b) Adrenaline
(c) Estrogen (d) Oxytocin

U.P.P.S.C. (GIC) 2010
U.P.P.C.S. (Spl.) (Mains) 2004
U.P.P.C.S. (Pre) 2003
U.P. U.D.A./L.D.A. (Pre) 2002
U.P. P.C.S. (Pre) 2001

Ans. (b)

Adrenaline (Epinephrine) is commonly known as the 'fight or flight hormone'. Adrenaline is a hormone released from the adrenal glands and its major action together with noradrenaline is to prepare the body for fight or flight. Adrenaline causes dilation of blood vessels (vasodilation) which supply the brain, skeletal muscles, heart, lungs, liver, adipose tissues, sensory organs etc. Due to increased blood supply, these organs become very active and excited inducing alarm reaction, contraction of cardiac muscles, increasing both rate and force of heartbeat, pulse rate, arterial pressure and cardiac output. It produces a feeling of excitement.

41. Secretion of which hormone increases heart beat and produces a feeling of excitement?

(a) Cortisone (b) Insulin
(c) Adrenaline (d) Testosterone

R.A.S./R.T.S.(Pre) 2003

Ans. (c)

See the explanation of above question.

42. By whom estrogen is produced –

(a) Egg (b) Follicles
(c) Corpus luteum (d) Uterus

R.A.S./R.T.S. (Pre) 1994

Ans. (b)

Estrogen hormones are produced primarily by the ovaries. They are released by the follicles on the ovaries and are also secreted by the corpus luteum after the egg has been released from the follicle and from the placenta. The primary function of estrogens is the development of female secondary sexual characteristics. These included breasts, endometrium, the menstrual cycle, etc.

43. What is estrogen?

(a) Bone (b) Hormone

(c) Enzyme (d) Vitamin

U.P.P.C.S. (Pre) 2000

U.P. Lower (Pre) 2004

Ans. (b)

See the explanation of above question.

44. The female sex hormone is:

(a) Estrogen (b) Androgen

(c) Insulin (d) Oxytocin

(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (a)

Among the given options, estrogen (oestrogen) is the female sex hormone. It is produced primarily by the ovaries and responsible for the development and function of the female reproductive organs and sexual characteristics.

45. Which one of the following is a female sex-hormone?

(a) Androsterone (b) Testosterone

(c) Estrone (d) Thyroxine

U.P. R.O./A.R.O. (Pre) 2017

Ans. (c)

Estrone is a weak postmenopausal estrogen and a minor female sex hormone. Estrogen (Oestrogen) is the actual female sex hormone. Testosterone is a male sex hormone which also includes Androgen. Thyroxine hormone is secreted the by thyroid gland which regulates the metabolism of the body.

46. After menopause, production of which of the following hormones does not take place in women ?

(a) Progesterone (b) Testosterone

(c) Estrogen (d) None of these

U.P.P.C.S. (Pre) 2014

Ans. (*)

Menopause is marked by the following hormonal changes: Ovarian secretion of estrogen and progesterone ends. Although the ovaries stop producing estrogen, they still continue to produce a small amount of the male hormone testosterone, which can be converted to estrogen (estradiol) in body fat. In addition, the adrenal gland continues to produce androstenedione (a male hormone), which is converted to estrone and estradiol in the body fat. The total estrogen produced after menopause is however far less than that produced during a woman's reproductive years. Initially, in the answer key released by UPPSC , the correct answer to this question was marked as option (c) but later in revised answer key, UPPSC has cancelled this question.

47. With reference to the human body, consider the following statements :

1. The production of somatotropin goes up when a person exercises.

2. Men's testes produce progesterone.

3. Women's adrenal glands secrete testosterone.

4. Stress causes the adrenals to release very less amount of cortisol than usual.

Which of these statements are correct?

(a) 1, 2, 3 and 4 (b) 1, 2 and 3

(c) 2, 3 and 4 (d) 1 and 4

I.A.S. (Pre) 2002

Ans. (b)

Growth hormone (GH), also known as somatotropin, is a peptide hormone that stimulates growth, cell production and regeneration in human and other animals. The production of somatotropin goes up when a person exercises. Progesterone is primarily a female sex hormone, but it is produced in small amount within men. Progesterone is produced in the ovaries, adrenal glands, testes and brain, and in the placenta during pregnancy. Testosterone is a naturally occuring sex hormone that is produced in a men's testicles. Small amount of testosterone is also produced in a woman's ovaries and adrenal glands. Cortisol is produced in humans by the zona fasciculata of the adrenal cortex within the adrenal gland. Its release is increased in response to stress and low blood-glucose concentration. Hence statements 1, 2 and 3 are correct, while statement 4 is incorrect.

48. Match the List-I with List-II and select the correct answer with codes given below :

List-I	**List-II**
A. Hormone	**1. Lipase**
B. Enzyme	**2. Testosterone**
C. Phospholipid	**3. Lecithin**
D. Polymer	**4. Polythene**

Code :

	A	B	C	D
(a)	2	1	3	4
(b)	4	1	2	3
(c)	2	3	4	1
(d)	1	2	3	4

U.P.P.C.S. (Pre) 1997

Ans. (a)

The correctly matched lists are as follows :

Hormone	–	Testosterone
Enzyme	–	Lipase
Phospholipid	–	Lecithin
Polymer	–	Polythene